

# SULAMAN KHAN

Senior Backend Engineer | Go • Python • Node.js • Distributed Systems

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github.com/sulaman0 • linkedin.com/in/suleman-khan-eng • stackoverflow.com/users/8344330  
Full overlap with Gulf hours • 3–4 hours with UK/EU • US mornings covered

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## PROFESSIONAL SUMMARY

Senior backend engineer with 8 years building systems that carry real load: a fintech platform that processed \$1M+ in its first quarter, a CMS serving 100,000+ daily active users, an API tuned to handle 1M+ requests, and a 1M-user Node.js service I migrated to Go to end recurring memory failures. I work in Go, Python and Node.js across gRPC and REST microservices, PostgreSQL, MySQL, Redis and Elasticsearch, deployed on AWS with Docker and Kubernetes. I have also built Layer 1 blockchain infrastructure and, more recently, production LLM systems — both of which are exercises in the same thing: correctness under load, observability, and graceful failure.

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## CORE SKILLS

**Languages:** Go, Python, Node.js, TypeScript, Rust, PHP/Laravel, Solidity

**Backend:** gRPC, Protocol Buffers, REST APIs, GraphQL, microservices, event-driven architecture, message queues, concurrency (goroutines, channels), system design

**Data:** PostgreSQL, MySQL, Redis, Elasticsearch, pgvector, query profiling and index optimization

**Cloud & DevOps:** AWS (Lambda, API Gateway), Docker, Kubernetes, Nginx, DigitalOcean, GitHub Actions, CI/CD, Datadog, observability

**Frontend:** React, React Native, Expo, TypeScript

**AI / LLM:** Python, FastAPI, LangChain, LangGraph, RAG pipelines, ChromaDB, pgvector, LLM agents, Model Context Protocol (MCP), OpenAI and Anthropic APIs

**Blockchain:** Substrate / Polkadot SDK, EVM compatibility (Frontier), Aura and PoS consensus, Solidity, load and gas-limit testing

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## PROFESSIONAL EXPERIENCE

### Backend & Blockchain Engineer → AI Engineer

Aug 2021 – Present

*Vaival Technologies • Client: Vulcan Forged (US) • Remote*

- **Go migration at scale:** Migrated a Node.js service serving 1M users to Go, resolving persistent memory issues that had required scheduled restarts; used goroutines and channels for the concurrent paths and gRPC for inter-service communication.
- **High-throughput APIs:** Optimized a blockchain art API to handle 1M+ requests through careful server-resource management and caching; built Go and Node.js services serving real-time on-chain data over GraphQL.
- **Distributed systems:** Built and operated Layer 1 blockchain infrastructure — block explorer, DEX, NFT marketplace, bridge and swap modules — including consensus configuration, validator-set management, load testing and gas-limit tuning.
- **Security-critical paths:** Implemented transaction signing, DeFi protocol integration and secure smart-contract interaction where a defect is publicly visible and cannot be quietly rolled back.
- **AI systems (Jan 2024–present):** Built production RAG pipelines and LLM agents as async FastAPI services with SSE streaming, multi-tenant retrieval isolation verified by security tests, per-request cost and latency observability, and multi-provider failover.

### Full Stack Engineer (Web & Mobile)

Jun 2018 – Dec 2021

*B4USoft Pvt. Ltd. • Lahore, Pakistan*

- **Fintech at scale:** Built secure transaction, authentication, fraud-prevention and data-security features; the launched application processed \$1M+ in transactions in its first quarter.
- **High-traffic systems:** Designed and optimized a CMS supporting 100,000+ daily active users; built React Native e-commerce applications backed by Go APIs with Stripe and PayPal payments and inventory management.
- **Cloud & cost:** Moved workloads to serverless on AWS Lambda with Docker-based deployment and Datadog monitoring, cutting operational costs 40% while improving scalability.
- **Technical leadership:** Led projects to CMMI 2.0 certification; raised an app store rating from 4.3 to 4.9 and reduced startup time to 10 seconds.

## SELECTED PROJECTS

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Note on code visibility: production work for client engagements lives in client-owned repositories. Public architecture write-ups for the systems below are published at [github.com/sulaman0](https://github.com/sulaman0).

- **ZATCA Phase 2 E-Invoicing** — Saudi Arabia tax-compliance implementation in PHP: Fatoora integration, compliant XML generation and certificate signing. Open-sourced; 13 stars and 7 forks, in use by other teams. Regulated-domain engineering where the specification admits no approximation.
- **Shopify Apps** — Built and maintain Cartly, Sales Pop-up and Wheelify, used by thousands of merchants — shipping inside another platform's constraints, review process and API limits.
- **Looqer — AI Shopify App** — Full monorepo shipped end to end: React and Polaris admin dashboard, Node/Express/Prisma API, embedded vanilla-JS widget runtime, Shopify theme extension. Deployed on DigitalOcean with Neon PostgreSQL, Nginx and Cloudflare, with tiered subscription billing.
- **Multi-Tenant RAG Service** — Production RAG API in Python and FastAPI with ChromaDB, SSE token streaming, per-request tenant isolation, and cost and latency observability middleware.
- **Substrate Chain & PoS Consensus** — Substrate-based chain with Aura consensus, validator-set and node-authorization pallets; separately, a proof-of-stake consensus implementation in Go with stake-weighted leader selection and block validation.

## SELECTED ACHIEVEMENTS

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- Migrated a 1M-user Node.js service to Go, eliminating recurring memory failures.
- Fintech application processed \$1M+ in transactions within its first quarter post-launch.
- Optimized an API to handle 1M+ requests; scaled a CMS to 100,000+ daily active users.
- Cut operational costs 40% through serverless architecture while improving scalability.
- Led projects to CMMI 2.0 certification; app rating 4.3 → 4.9; startup time cut to 10 seconds.

## EDUCATION

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### B.S. Information Technology

2014 – 2018

*University of the Punjab, Lahore, Pakistan*

- Applied Concurrency in Go — goroutines, channels, concurrent systems design (2023)